FIREWORKS DEPLOY BUILDATHON · NOV 2026

Thought → Device

An open LLM cannot read brain signals, even fine-tuned. Given a classic decoder as a tool, it controls an exoskeleton and a robot: 52.8% → 62.5% on new people.

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All numbers measured on 144 held-out trials · 4 people

THE PROBLEM

A person who cannot move can steer a machine by imagining a hand movement.

Who needs it

People with ALS, spinal cord injury or stroke.

What blocks it

Each new user records a calibration session before the decoder works for them.

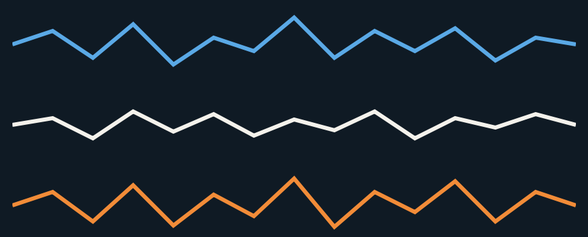
Our question

With no calibration, can a fine-tuned open LLM match the classic decoder?

THE TASK

Four seconds of 64-channel EEG become six numbers of text.

1 · RAW EEG



64 channels, 8–30 Hz, 160 Hz

→

2 · CSP FILTERS → TEXT

```
component 1: +1.34
component 2: +1.15
component 3: +1.28
component 4: +1.12
component 5: +1.42
component 6: +1.38
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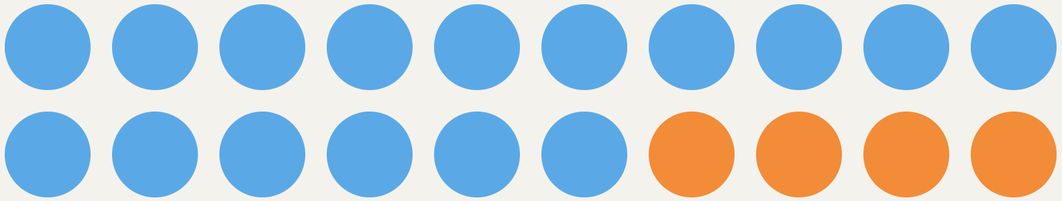
→

3 · ANSWER

LEFT

THE DATA

Public recordings. The test people are never seen in training.



16 people · 576 training trials (S001–S016)

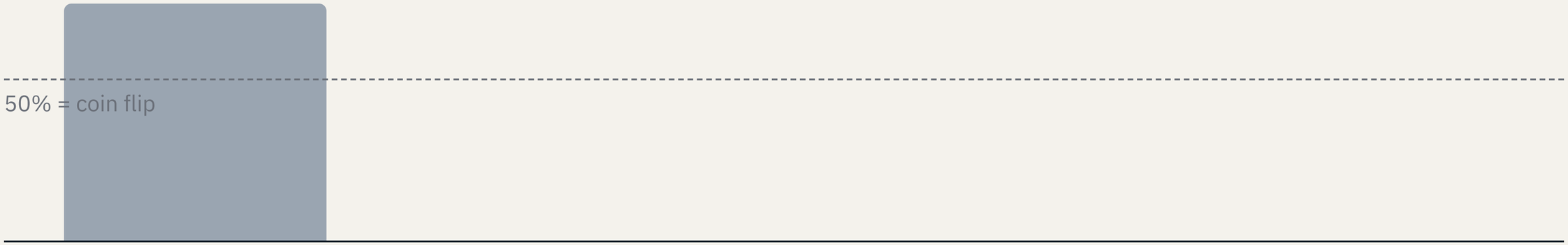
4 people · 144 test trials (S101–S104)

PhysioNet EEG Motor Movement/Imagery (Schalk et al.). Imagined left or right fist, runs 4, 8, 12. Half left, half right.

Source: huggingface.co/datasets/raei/Schalk-2009-EEGMotorMovementImagery (ODC-By)

BEFORE

Off the shelf, the LLM says “LEFT” 85% of the time.



GLM 5.3 Flash
base · **52.8%**

76 of 144 correct · 95% CI 44–61% · not above chance (p = 0.28). Answered LEFT on 122 of 144 trials.

Same prompt and 144 trials for every system · thinking model, reasoning effort low, temperature 0

WHAT WE DID

LoRA fine-tune of GLM 5.3 Flash on Fireworks.

STEP 1
CSP features
fit on 16 people

→

STEP 2
JSONL
576 train lines

→

STEP 3
LoRA SFT
rank 8, 3 epochs

→

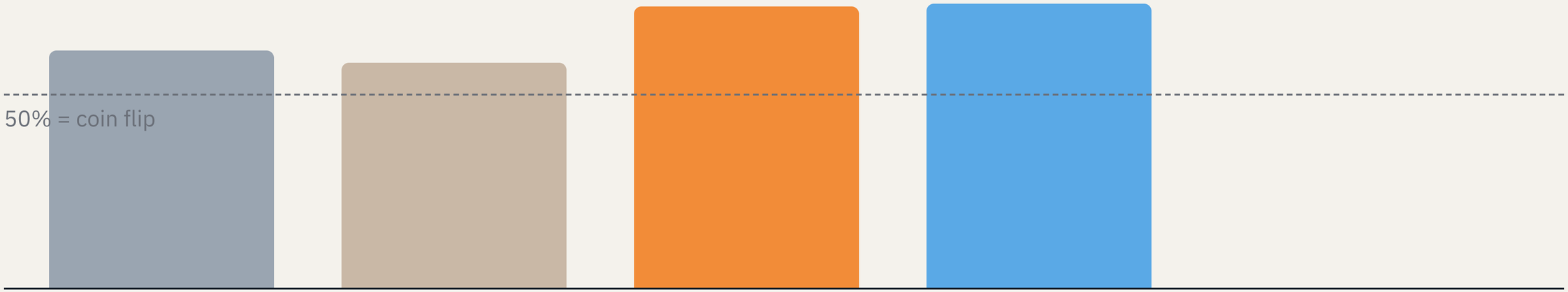
STEP 4
Serverless LoRA
same prompt

14 min **$2.54**

Fireworks job tjf37e7j · answer-only targets · 14 min from submit to done · cost = Fireworks estimatedCost on the job

BEFORE → AFTER

Fine-tuning failed. A harness took the LLM from 52.8% to 62.5%.



Base LLM
52.8% · ~0.6 s

LLM + LoRA
50.0–51.4% · ~0.5 s

LLM + decoder tool
62.5% · ~0.6 s

CSP + LDA alone
63.2% · <0.1 ms

144 held-out trials, 4 unseen people. McNemar vs base: harness p = 0.065 (borderline) · LDA p = 0.044 · LoRA p = 0.87. Training cost, 3 jobs: $9.22.

EVERY RUN

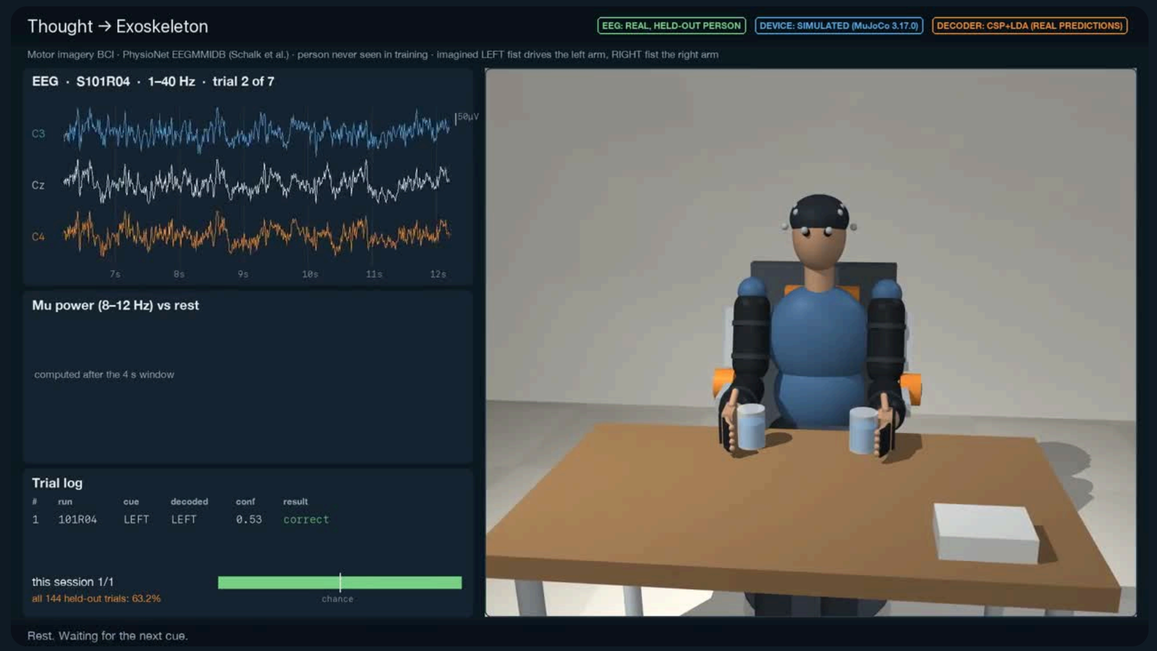
Every run, including the failures.

Run (test = 144 trials, 4 people)	Accuracy	Status
Band-power change vs rest, 7 electrodes · logistic reg.	47.2%	discarded, below chance
Left–right laterality, 5 pairs · logistic reg.	54.2%	discarded (61.3% in train CV)
CSP + LDA on raw EEG	63.2%	reference decoder
GLM 5.3 Flash, off the shelf, CSP text	52.8%	before · LEFT 85%
GLM 5.3 Flash + LoRA (rank 8, 3 epochs) · $2.54	50.0–51.4%	no gain, LEFT ~80%
Distill LoRAs: LDA labels ($2.65), LDA arithmetic ($4.03)	—	trained, not scored
Base LLM + decoder score pasted in prompt	56.9%	ignores score often
Harness: base LLM calls CSP+LDA as a tool	**62.5%**	**best LLM system, no training**

Disclosure: the test people were scored for the two failed formats before CSP was chosen. Clean next step: confirm on 4 new people (S105–S108).

DEMO · RECORDED EEG → SIMULATED DEVICES

Real decoder output drives an exoskeleton and a fetch robot.



Exoskeleton. Person S101, first 7 trials in order: 3 of 7 correct. Wrong decodes move the wrong arm.



Assistant robot. Brain picks the goal, robot plans the motion. S103: water in 3 commands.

Decoder: CSP+LDA, real per-trial predictions (63.2%). Same menu, 4 test people: S103 water in 3 · S102 in 9 · S104 confirmed Apple by mistake · S101 no selection in 14.

NEXT

Next: let the LLM agent handle low-confidence decodes, then test on 4 fresh people.

Repo
github.com/renantrendt/fireworks-anonymize/eeg-bci

Training
3 Fireworks jobs · $9.22

Data
PhysioNet EEGMMIDB